

Black Hole Horizons as Thermodynamic Encoding Surfaces: Throughput, Information Flow, and the Two-Horizon Framework

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

We apply the horizon thermodynamics framework of the Informational Actualization Model (IAM) to black hole horizons, extending the Jacobson–Cai–Kim procedure from cosmological to local horizons. Three principal results emerge. (1) The maximum encoding throughput of a Schwarzschild black hole horizon is $\Gamma_{\text{thermal}} = c^3/(1920 GM \ln 2) \text{ bits s}^{-1}$, derived as the ratio of Hawking luminosity to the Landauer cost per bit, scaling inversely with mass. (2) The Stefan-Boltzmann law applied to the black hole horizon at the Hawking temperature exactly reproduces the Hawking luminosity, $P_{\text{SB}} = P_{\text{Hawking}}$, for all masses — establishing that Hawking radiation is Stefan-Boltzmann radiation from a thermal encoding surface. (3) The net radiative power between the black hole horizon (at T_{BH}) and the cosmological apparent horizon (at the Gibbons-Hawking temperature T_{GH}) is $P_{\text{net}} = \sigma_{\text{SB}} A_{\text{BH}} (T_{\text{BH}}^4 - T_{\text{GH}}^4)$, with thermal equilibrium mass $M_{\text{eq}} = c^3/(4GH)$ exceeding all astrophysical black holes at every cosmological epoch by many orders of magnitude. All real black holes radiate; information flows monotonically from local hot horizons to the global cold horizon. The information paradox dissolves as ordinary thermodynamic flow between encoding surfaces. We derive the holographic saturation condition for black hole formation and establish its equivalence to Thorne’s hoop conjecture in informational language.

1 Introduction

The thermodynamic nature of black hole horizons has been recognized since the foundational work of Bekenstein Bekenstein1973 and Hawking Hawking1975. The Bekenstein-Hawking entropy $S = A/(4\ell_P^2)$ and the Hawking temperature $T = \hbar c^3/(8\pi GM k_B)$ establish black holes as thermal objects. Jacobson Jacobson1995 derived the Einstein field equations from horizon thermodynamics, showing that gravity is an equation of state.

Cai & Kim CaiKim2005 extended this to FRW spacetimes, demonstrating that the Friedmann equations follow from the first law applied to the apparent horizon.

The Informational Actualization Model (IAM) Mahaffey2026a,Mahaffey2026b extends this program by identifying gravitational decoherence as a source of additional informational entropy that, when encoded on the cosmological apparent horizon via the Cai-Kim procedure, modifies the matter-sector perturbation equations to produce a dual-sector structure ($\mu < 1$ for matter, $\Sigma = 1$ for photons) while leaving the background expansion as standard Λ CDM.

In this paper, we apply the same thermodynamic framework to local black hole horizons. If the cosmic horizon is an encoding surface governed by horizon thermodynamics, then black hole horizons must be too — they are the most efficient encoding surfaces in nature. We derive the encoding throughput, establish the exact identity between Stefan-Boltzmann and Hawking radiation, construct the two-horizon thermodynamic framework governing information flow in an IAM universe, and derive the holographic saturation condition for black hole formation.

2 Encoding Throughput of a Black Hole Horizon

2.1 Thermally-limited encoding rate

The maximum rate at which a black hole horizon can encode information is set by the ratio of the available radiated power to the Landauer cost per bit:

$$\Gamma_{\text{thermal}} = \frac{P}{k_B T_{\text{BH}} \ln 2} \quad (1)$$

Using the Hawking luminosity Hawking1975 $P = \hbar c^6 / (15360\pi G^2 M^2)$ and the Hawking temperature $T_{\text{BH}} = \hbar c^3 / (8\pi G M k_B)$:

$$\Gamma_{\text{thermal}} = \frac{\hbar c^6 / (15360\pi G^2 M^2)}{k_B \cdot \hbar c^3 / (8\pi G M k_B) \cdot \ln 2} = \frac{c^3}{1920 G M \ln 2} \quad (2)$$

This result is exact for a Schwarzschild black hole. The throughput scales inversely with mass: smaller, hotter horizons encode faster.

2.2 Numerical values

Table 1: Encoding throughput, Bekenstein-Hawking entropy (in nats, $S_{\text{BH}} = 4\pi G M^2 / \hbar c$), and evaporation time $\tau = 5120\pi G^2 M^3 / (\hbar c^4)$ for representative black hole masses. All values computed with exact CODATA 2018 constants.

Mass ($M_\odot$)	Γ (bits s ⁻¹)	S_{BH} (nats)	τ_{evap} (yr)
1	1.525×10^2	1.049×10^{77}	2.10×10^{67}
10	1.525×10^1	1.049×10^{79}	2.10×10^{70}
10^6	1.525×10^{-4}	1.049×10^{89}	2.10×10^{85}
10^9	1.525×10^{-7}	1.049×10^{95}	2.10×10^{94}

The evaporation time τ_{evap} represents the time to transfer all locally-encoded information to the cosmic horizon via Hawking radiation.

2.3 Physical interpretation

Γ_{thermal} is the maximum rate at which the horizon processes information — both encoding new information from incoming gravitational decoherence and radiating existing information outward via Hawking emission. Section 3 establishes that these two rates are identical.

3 Stefan-Boltzmann–Hawking Identity

3.1 Exact identity

The Stefan-Boltzmann law applied to the black hole horizon:

$$P_{\text{SB}} = \sigma_{\text{SB}} A_{\text{BH}} T_{\text{BH}}^4 \quad (3)$$

with $\sigma_{\text{SB}} = \pi^2 k_B^4 / (60 \hbar^3 c^2)$, $A_{\text{BH}} = 16\pi G^2 M^2 / c^4$, and $T_{\text{BH}} = \hbar c^3 / (8\pi G M k_B)$. Substituting:

$$\begin{aligned} P_{\text{SB}} &= \frac{\pi^2 k_B^4}{60 \hbar^3 c^2} \cdot \frac{16\pi G^2 M^2}{c^4} \cdot \left(\frac{\hbar c^3}{8\pi G M k_B} \right)^4 \\ &= \frac{\pi^2 k_B^4}{60 \hbar^3 c^2} \cdot \frac{16\pi G^2 M^2}{c^4} \cdot \frac{\hbar^4 c^{12}}{4096 \pi^4 G^4 M^4 k_B^4} \\ &= \frac{\hbar c^6}{15360 \pi G^2 M^2} \end{aligned} \quad (4)$$

This is identically the Hawking luminosity:

$$\boxed{\frac{P_{\text{SB}}}{P_{\text{Hawking}}} = 1} \quad (5)$$

for all masses, verified numerically to 12 significant figures using exact CODATA 2018 constants.

3.2 Interpretation

This identity is not a consistency check — it is a physical statement. Hawking radiation *is* Stefan-Boltzmann radiation from a thermal surface at the Hawking temperature. The horizon radiates because it is hot, and it is hot because encoding information at the Landauer cost $k_B T \ln 2$ per bit releases energy at temperature T . The microscopic derivation (QFT in curved spacetime) and the macroscopic thermodynamic description agree exactly, confirming that the horizon’s thermodynamic properties are physical, not analogical.

3.3 The Hawking bit rate

The rate at which information departs the black hole via Hawking radiation is:

$$\frac{dS_{\text{transfer}}}{dt} = \frac{P_{\text{Hawking}}}{k_B T_{\text{BH}} \ln 2} = \frac{c^3}{1920 G M \ln 2} = \Gamma_{\text{thermal}} \quad (6)$$

The encoding throughput (Eq. 2) and the Hawking emission rate are the same quantity viewed from two directions. The horizon encodes at precisely the rate it radiates.

4 The Two-Horizon Framework

4.1 Cosmic horizon temperature

The cosmological apparent horizon at $\tilde{r}_A = c/H$ has the Gibbons-Hawking temperature GibbonsHawking1977:

$$T_{\text{GH}} = \frac{\hbar H}{2\pi k_B} \quad (7)$$

At the present epoch ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$): $T_{\text{GH}} = 2.66 \times 10^{-30} \text{ K}$, many orders of magnitude below any black hole Hawking temperature.

4.2 Net radiation between two thermal horizons

Both the black hole horizon and the cosmic horizon are thermal surfaces. The net radiative power between them is:

$$P_{\text{net}} = \sigma_{\text{SB}} A_{\text{BH}} (T_{\text{BH}}^4 - T_{\text{GH}}^4) \quad (8)$$

Three regimes exist: (1) $T_{\text{BH}} > T_{\text{GH}}$ ($M < M_{\text{eq}}$): standard Hawking radiation, information flows $\text{BH} \rightarrow \text{cosmic horizon}$. (2) $T_{\text{BH}} = T_{\text{GH}}$ ($M = M_{\text{eq}}$): thermal equilibrium, no net flow. (3) $T_{\text{BH}} < T_{\text{GH}}$ ($M > M_{\text{eq}}$): net absorption from cosmic horizon.

4.3 Equilibrium mass

Setting $T_{\text{BH}} = T_{\text{GH}}$:

$$\frac{\hbar c^3}{8\pi G M_{\text{eq}} k_B} = \frac{\hbar H}{2\pi k_B} \Rightarrow M_{\text{eq}} = \frac{c^3}{4GH} \quad (9)$$

Table 2: Thermal equilibrium mass $M_{\text{eq}} = c^3/(4GH)$ and Gibbons-Hawking temperature T_{GH} at representative cosmological epochs. Hubble parameter computed from $H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda + \Omega_r(1+z)^4}$ with Planck 2018 parameters. The largest known black hole (TON 618, $6.6 \times 10^{10} M_\odot$) is far below M_{eq} at all epochs shown.

Epoch	$H \text{ (s}^{-1}\text{)}$	$T_{\text{GH}} \text{ (K)}$	$M_{\text{eq}} \text{ (} M_\odot \text{)}$
$z = 0$	2.18×10^{-18}	2.66×10^{-30}	2.32×10^{22}
$z = 1$	3.91×10^{-18}	4.76×10^{-30}	1.30×10^{22}
$z = 10^3$	4.41×10^{-14}	5.36×10^{-26}	1.15×10^{18}
$z = 10^6$	2.09×10^{-8}	2.54×10^{-20}	2.43×10^{12}
$z = 10^{10}$	2.08×10^0	2.53×10^{-12}	2.44×10^4

All astrophysical black holes at all cosmological epochs shown satisfy $M \ll M_{\text{eq}}$. The cosmic bath correction $(T_{\text{GH}}/T_{\text{BH}})^4 \approx 3.4 \times 10^{-90}$ for a solar-mass black hole today. Hawking radiation proceeds at the full uncorrected rate to extraordinary precision.

4.4 Information flow direction

The thermodynamic flow is unambiguous: information encoded on local (hot) black hole horizons radiates outward toward the global (cold) cosmic horizon. The sequence is: (1) gravitational decoherence produces information locally; (2) the information is encoded on the nearest available horizon — the black hole; (3) the black hole radiates at T_{BH} , transferring information to the bulk; (4) the information is ultimately re-encoded on the cosmic horizon at T_{GH} . The timescales for step (3) are $\sim 10^{67}$ yr and above (Table 1), so astrophysical black holes are effectively permanent local storage on cosmological timescales.

5 Resolution of the Information Paradox

5.1 The standard paradox

Hawking’s original calculation Hawking1975 suggested that information falling into a black hole is irretrievably lost upon evaporation, violating unitarity. The paradox arises from treating the black hole horizon as a one-way membrane that destroys information.

5.2 Resolution in the two-horizon framework

In IAM, the paradox dissolves on three grounds. First, information is not destroyed — it is transferred from a hot local encoding surface (BH horizon) to a cooler global encoding surface (cosmic horizon), exactly as heat transfers from a hot body to a cold reservoir. Second, the transfer obeys the second law: energy and information flow from hot to cold at the thermally-limited rate Γ_{thermal} . Third, the cosmic horizon is the final repository — all locally-encoded information migrates to the global ledger as the universe approaches thermodynamic equilibrium.

The resolution is not that information escapes through subtle correlations in Hawking radiation Page1993. It is that information was never trapped: it sits on a thermal surface that radiates by ordinary Stefan-Boltzmann thermodynamics, transferring its contents to a cooler surface at the rate set by the temperature difference.

6 Black Hole Formation: The Holographic Saturation Condition

6.1 Saturation as the formation criterion

In IAM, a black hole forms when the local information production from gravitational decoherence saturates the holographic bound on the bounding surface:

$$\frac{S_{\text{info}}(R)}{A(R)} \geq \frac{1}{4\ell_P^2} \quad (10)$$

where S_{info} is the information produced by decoherence within radius R and $A = 4\pi R^2$ is the bounding surface area.

6.2 Equivalence to the hoop conjecture

Thorne’s hoop conjecture Thorne1972 states that a black hole forms when mass M is compressed into a region with circumference $C \leq 4\pi GM/c^2$. The IAM saturation condition with $S_{\text{info}} = S_{\text{BH}} = 4\pi GM^2/(\hbar c)$ at the Schwarzschild radius $R_s = 2GM/c^2$ gives:

$$\frac{4\pi GM^2/(\hbar c)}{4\pi \cdot 4G^2 M^2/c^4} = \frac{c^3}{4\hbar G} = \frac{1}{4\ell_P^2} \quad (11)$$

The conditions are identically satisfied. The hoop conjecture is the holographic bound expressed in geometric language; IAM’s saturation condition is the hoop conjecture expressed in informational language. The same mathematics, with the thermodynamic *reason* made explicit: a black hole forms when the local information production rate saturates the surface’s encoding capacity.

6.3 Early universe: BH-dominated encoding

IAM’s activation function $E(a) = \exp(1 - 1/a)$ tracks the cosmic horizon’s informational entropy. At early times ($E(a) \approx 0$), information is encoded primarily on local black hole horizons, not on the small cosmic horizon. As the universe expands and large-scale structure formation produces distributed decoherence, the cosmic horizon grows and takes over as the dominant encoder.

This encoding transition provides a natural explanation for several observations: rapid early black hole formation is thermodynamically required — the early universe needs encoding surfaces, and black holes are the most efficient ones available before the cosmic horizon grows; the observed population of massive early black holes in JWST data is expected, not anomalous Mahaffey2026c; and the M - σ relation emerges as each galaxy’s SMBH is sized to the galaxy’s local information production rate Mahaffey2026c.

6.4 Primordial black holes

During radiation domination, electromagnetic interactions decohere particles far faster than gravity acts. Gravitational decoherence is subdominant. IAM therefore does not modify the primordial BH formation threshold $\delta_c \approx 1/3$ — primordial BHs form by the standard Carr mechanism Carr1975. IAM’s contribution to structure formation begins after recombination ($z \sim 1100$), when photon decoupling makes gravitational decoherence the dominant irreversible information source.

7 Seed Black Hole Masses from the Information Budget

After recombination, for collapsing halos at $z \sim 20$ – 30 , the black hole mass required to encode the local information budget is:

$$M_{\text{BH}} = \frac{m_{\text{Pl}}}{2\sqrt{\pi}} \sqrt{S_{\text{collapse}}} \quad (12)$$

where S_{collapse} (in nats) is the total irreversible gravitational information produced during collapse and $m_{\text{Pl}} = \sqrt{\hbar c/G}$ is the Planck mass.

Table 3: Predicted seed black hole masses from the information budget of gravitational collapse. The mass is set by the information content of the collapse, not by accretion history, providing a formation channel for the massive early black holes observed by JWST.

S_{collapse} (nats)	M_{BH} ($M_{\odot}$)	Classification
10^{77}	~ 1	Stellar mass
10^{83}	$\sim 10^3$	Intermediate mass
10^{89}	$\sim 10^6$	Supermassive
10^{95}	$\sim 10^9$	Ultra-massive

The mass scales with $\sqrt{S_{\text{collapse}}}$, so each three-order-of-magnitude increase in information content produces a three-order-of-magnitude increase in seed mass. This provides a formation channel for JWST’s massive early black holes that requires neither super-Eddington accretion nor exotic seed mechanisms — only the information budget of the collapse.

8 Discussion

8.1 Two regimes of one mechanism

IAM’s feedback loop — decoherence $\rightarrow$ information production $\rightarrow$ Landauer cost $\rightarrow$ horizon encoding $\rightarrow$ modified expansion — operates in two regimes. In the cosmological (self-regulating) regime, distributed decoherence produces information gradually, encoded on the growing cosmic horizon. Expansion dilutes matter, slowing further collapse. The feedback reaches equilibrium, producing $\mu < 1$, $\Sigma = 1$. In the collapse (runaway) regime, concentrated decoherence produces information explosively in a contracting region. Local encoding capacity saturates. The feedback runs away, producing a new horizon. Same physics, different boundary conditions.

8.2 Connection to the M – σ relation

The $M_{\text{BH}} \propto \sigma^4$ relation derived in the companion paper Mahaffey2026c is the steady-state limit of the seed mass formula: instead of a single collapse event, the SMBH accumulates information from the galaxy’s ongoing gravitational decoherence over the Hubble time, with the irreversible fraction scaling as v^2/c^2 from the post-Newtonian radiative order. The normalization contains Ω_m and H_0 , connecting black hole masses directly to the expansion history.

8.3 Testable predictions

9 Conclusion

Black hole horizons treated as thermal encoding surfaces within IAM’s horizon thermodynamics framework yield three exact results: the encoding throughput $\Gamma_{\text{thermal}} = c^3/(1920 GM \ln 2)$, the Stefan-Boltzmann–Hawking identity $P_{\text{SB}} = P_{\text{Hawking}}$, and the monotonic information flow from local hot horizons to the global cold cosmic horizon.

Table 4: Testable predictions of the two-horizon IAM framework.

Prediction	Value	Test
Encoding throughput	$\Gamma = c^3/(1920 GM \ln 2)$	Constrains BH information processing rate
$P_{\text{SB}} = P_{\text{Hawking}}$	Exact identity (Eq. 5)	Established analytically
All BHs radiate at all epochs	$M_{\text{eq}} > 10^{22} M_{\odot}$ at $z = 0$	Consistent with no BH absorption observed
No PBH formation modification	δ_c unchanged	Consistent with PBH constraints
Early massive BHs via information budget	Table 3	Testable with JWST
$M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$	15.7% at $z = 0$, redshift-dependent	eROSITA/Euclid clusters Mahaffey2026d

The thermal equilibrium mass $M_{\text{eq}} = c^3/(4GH)$ exceeds all astrophysical black holes at every cosmological epoch, confirming that all real black holes radiate at the full Hawking rate throughout cosmic history.

The information paradox dissolves as ordinary thermodynamic flow: information is not destroyed, it migrates from a hot local surface to a cool global one at the thermally-limited Hawking rate. The hoop conjecture is the holographic bound in geometric language; the saturation condition is the hoop conjecture in informational language. The same mechanism that modifies cosmic expansion ($\mu < 1$, $\Sigma = 1$) creates the strongest gravitational objects in the universe, differing only in boundary conditions. One mechanism, two regimes, every scale.

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